Cody Melcher

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Statistics and Data Science GIDP, University of Arizona, Tucson, AZ 85721

EDUCATION

University of Arizona May 2026 (expected)
Tucson, AZ
PhD in Statistics & Data Science; PhD Minor in Systems and Industrial Engineering

University of Arizona December 2019
Tucson, AZ
M.S. in Economics

Reed College May 2014
Portland, OR
B.A. in Economics

RESEARCH INTERESTS

Min–max optimization, first-order algorithms, robust learning, and deployment of optimization algorithms on embedded devices.

RESEARCH AND WORK EXPERIENCE

Research Assistant, University of Arizona (Dr. Erfan Yazdandoost Hamedani) Nov 2023 – Present
Tucson, AZ

- **Linear convergence without strong convexity:** Introduced two-sided Quadratic Functional Growth (2-QFG) and two-sided Quadratic Gradient Growth (2-QGG) as curvature conditions strictly weaker than strong convexity that still guarantee linear convergence of first-order methods.
- **Semi-infinite nonconvex constrained optimization:** Developed an inexact dynamic barrier primal–dual (iDB–PD) method for constrained semi-infinite min–max problems that drives constraint violation toward feasibility without penalty parameters and guarantees convergence to an ε-KKT point under a mild Lojasiewicz-type regularity condition.
- **On-device optimization for embedded learning (TinyML):** Deployed deep learning models onto the Arduino Nano 33 BLE Sense for binary symbol recognition, MNIST, and Multi-MNIST classification, and extending iDB–PD for real-time constrained learning on-device under strict memory and precision limits.

TEACHING EXPERIENCE

Graduate Teaching Assistant, University of Arizona
Tucson, AZ

- **MATH 263: Introduction to Statistics and Biostatistics** Jan 2025-May 2025
  Led weekly recitations, held office hours, assisted with creating homework and quiz questions, and graded materials.
- **MATH 112: College Algebra** Aug 2022 – May 2023
  Created and graded homework, led review sessions, held office hours, and supported curriculum development. Awarded *Honorable Mention for Mathematics Graduate TA* (Spring 2023).

AWARDS

Carter Travel Award, University of Arizona Oct 2024
$600 travel award for research presentation.

Papers and Presentations

- **NeurIPS (Conference on Neural Information Processing Systems)** Dec 2025
  San Diego, CA
  Poster presentation of *Semi-infinite Nonconvex Constrained Min–Max Optimization.*

- **INFORMS Annual Meeting** Oct 2025
  Atlanta, GA
  Talk: *Semi-infinite Nonconvex Constrained Min–Max Optimization*, presented in the *Structured and Hierarchical Optimization* session.

- **INFORMS Annual Meeting** Oct 2024
  Seattle, WA
  Talk: *Incorporating Quadratic Functional Growth in Saddle Point Problems*, presented in the *Game Theory and Optimization: Minimax and Variational Inequality Approaches* session.

WORKING PAPERS

- C. Melcher, Z. Alizadeh, L. Hiett, A. Jalilzadeh, and E. Yazdandoost Hamedani, "Semi-infinite Nonconvex Constrained Min–Max Optimization," arXiv preprint arXiv:2510.12007, 2025.

- C. Melcher, A. Jalilzadeh, and E. Yazdandoost Hamedani, "Linear Convergence of a Unified Primal–Dual Algorithm for Convex–Concave Saddle Point Problems with Quadratic Growth," arXiv preprint arXiv:2510.11990, 2025.

LANGUAGES AND SKILLS

Programming: Python (PyTorch, Tensorflow), R, MATLAB, LaTeX, Stata